Integrating Large Language Models and Retrieval-Augmented Generation for Telepresence in Socially Assistive Robots

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***Abstract*—The global healthcare sector faces mounting pressures from aging populations and clinical staff shortages, necessitating infection-aware models of care. Socially assistive robots like Pepper offer natural, engaging interactions but are often constrained by legacy hardware and computational limits. This paper presents a novel architecture that integrates modern Large Language Models (LLMs) and Retrieval-Augmented Generation (RAG) with the Pepper robotic platform to overcome its native interoperability gaps. We propose a three-tier system comprising a legacy Python 2.7 robot interface layer, a Python 3.x backend server utilizing Whisper AI for speech-to-text, and a web-based user interface. By executing semantic search against a FAISS vector store of structured hospital data, the system grounds the Gemini 2.5 Flash model's responses, minimizing AI hallucinations and ensuring medically accurate administrative support. Real-world testing demonstrated an end-to-end artificial intelligence latency of 2.23 seconds and an optimized Word Error Rate (WER) of 9.97%. The system successfully bridged legacy NAOqi APIs with modern cloud infrastructure, achieving a 100% API success rate. These results validate the deployment of LLM-enhanced humanoid robots for patient triage, significantly improving interaction success and system usability in clinical environments.**

***Index Terms*—Socially Assistive Robots, Human-Robot Interaction, Large Language Models, Retrieval-Augmented Generation, Medical Robotics.**

I. Introduction

The global healthcare sector is undergoing a major transformation due to the combined pressures of an aging population, shortages in clinical staff, and the ongoing need for safer, infection-aware models of care. These challenges have increased the demand for intelligent systems that can support healthcare workers, improve service efficiency, and enhance patient experience.

Among the available social robots, Pepper was selected for this project because of its user-friendly and human-centered design. Unlike traditional tablet-based interfaces, Pepper provides a more natural and engaging interaction through speech, gesture, and physical presence. Its humanoid embodiment makes communication more approachable and can create greater comfort and positive emotional response, which is especially valuable in healthcare settings.

The importance of such systems is also reflected in the rapid growth of the medical robotics market, which is projected to increase from approximately USD 16.6 billion in 2023 to over USD 63.8 billion by 2032. This growth highlights the increasing role of robotics in the future of healthcare delivery.

Several healthcare robots have already contributed to this field. CareDo demonstrated the value of telepresence and infection-free care by being deployed in COVID-19 isolation

wards for remote patient checks. YuMi supported this system by performing physical tasks such as delivering medicine and food. ARI showed how integrating Large Language Models into social robots can improve usability and interaction success, especially in geriatric care. NAO has been used in human-robot interaction studies as a benchmark for emotional impact and engagement, while Furhat has contributed insights into conversational strategies and user feedback. Building on these developments, Pepper advances healthcare robotics by combining humanoid embodiment, patient-friendly interaction, and cloud-based generative AI in a unified medical assistant platform.

In the proposed system, user speech is first transcribed in approximately 0.5 seconds, after which the query is processed by the Gemini model with an average AI latency of 2.23 seconds, ranging from 1.43 seconds for the fastest response to 2.88 seconds for the slowest. This produced a fully reliable API success rate of 100% during testing, while supporting an overall conversational response time suitable for near-real-time patient interaction. These results indicate that the Pepper Medical Assistant is not only conceptually relevant to modern healthcare needs, but also technically capable of delivering responsive and dependable intelligent assistance in a hospital environment.

II. Literature Review

Table I summarizes the strongest findings extracted from the reviewed literature. The selected studies collectively show that socially assistive robots can improve patient engagement, emotional wellbeing, and interaction quality in healthcare settings. Among the reviewed works, the most significant quantitative gains were reported in studies integrating Large Language Models into social robots, where interaction success, usability, and acceptability improved substantially. Pepper-specific studies further demonstrate the robot's strong social presence, high engagement rates, and suitability for long-term care, pediatrics, and patient-facing assistance.

The literature indicates that the greatest value of socially assistive robots lies in their ability to enhance communication, engagement, and emotional comfort rather than replace medical professionals [1], [2]. In particular, LLM-enhanced robots show a major improvement over earlier scripted systems, suggesting that conversational intelligence is a key factor in successful deployment [7]. At the same time, implementation studies emphasize that real-world healthcare use also depends on reliability, latency, privacy, and staff acceptance [3]. These findings directly support the design of the Pepper Medical Assistant, which combines a user-friendly humanoid platform with an external AI-driven backend to improve interaction quality while addressing known limitations of onboard robot processing.

III. Problem Analysis and Requirements

The Pepper Medical Assistant is designed for deployment in a real hospital environment, where technical limitations, noisy surroundings, and data sensitivity create important design challenges.

One major problem is the interoperability gap between Pepper's legacy NAOqi SDK and modern AI frameworks. Pepper was originally built for structured and simple interactions, while current AI models require flexible processing and higher computational power. To solve this, the system follows an API-first approach in which Pepper handles user interaction, while speech processing, AI generation, and data access are managed by an external backend.

Another challenge is audio quality in hospital settings. Background noise can reduce speech-recognition accuracy, and Pepper's built-in recording does not provide advanced noise suppression [8]. Therefore, the system relies on backend-based speech processing and denoising, while the interface provides visual feedback such as "Listening" and "Thinking" to reduce user confusion and overlapping speech.

A third requirement is ethical compliance and data security. Since the system may handle patient-related information, access to sensitive data must be restricted. For this reason, the platform applies role-based access control (RBAC) and ensures that the AI assistant operates within strict boundaries, presenting Pepper as a support tool rather than a diagnostic authority.

Based on these challenges, the system must support voice interaction, information retrieval, and AI-based response generation, while also meeting key non-functional requirements such as low latency, usability, reliability, privacy, and scalability.

IV. Research Methodology (Overall System Architecture)

The Pepper Medical Assistant utilizes a sophisticated, multi-layered architecture designed to bridge the gap between legacy robotic hardware and modern generative artificial intelligence. The system is structured into three primary tiers: the Robot Interface Layer, the Backend Server Layer, and the User Interface Layer. This hierarchical framework ensures modularity, allowing each component to handle specific tasks ranging from low-level hardware control to high-level semantic reasoning.

The high-level architecture follows a centralized "Brain" model where the Pepper robot acts as the physical interface for data collection and user engagement, while an external backend server manages all heavy computational tasks, including speech-to-text (STT), natural language understanding (NLU), and knowledge retrieval [9]. This design specifically addresses the computational limitations of the robot's onboard system while maximizing the speed and accuracy of the AI responses.

The first tier, the Pepper Robot Layer, resides locally on the robot hardware and is governed by the NAOqi Framework, operating on a legacy version of Linux using Python 2.7. This layer manages sensory inputs and physical outputs, including speech capture, gesture control, and the tablet display. The robot utilizes its head-mounted 4-microphone array to record audio at a sampling rate of 16,000 Hz in a mono channel,

TABLE I
SYNTHESIS OF RECENT FINDINGS IN SOCIALLY ASSISTIVE ROBOTICS AND LLM INTEGRATION

Theme	Study / Paper	Setting / Focus	Key Findings & Implications for this Project
SARs in Healthcare	Nichol et al. (2024) [1]	Health & wellbeing	Strongest reported benefits in social interaction, mood, and loneliness. Supports using Pepper for emotional support, validating its behavioral outcome potential.
Assistive Support Robots	Giansanti et al. (2025) [2]	Healthcare, Therapy	Shows SARs help reduce depression and anxiety. Strengthens the argument that social robots can support patient experience in long-term care.
Pepper in Care	Blindheim et al. (2023) [5]	Elderly facilities	Created novel forms of activity beyond daily routine. Justifies Pepper as an already-validated platform capable of socially engaging assistance.
Pediatric Interaction	Rõstõsinskaja et al. (2025) [6]	Child-robot interaction	Children engaged quickly (2.0s initial engagement); 93.8% maintained eye contact. Provides strong evidence for Pepper's user-friendly embodiment.
LLMs for HRI	Blavette et al. (2025) [7]	Geriatric hospital	Error-free interactions increased from 27.8% to 70.2%; interaction success rose to 74.5%. Directly supports the use of LLM-enhanced dialogue in our architecture.
User Acceptance	Blavette et al. (2025) [7]	Geriatric hospital	Acceptability E-Scale improved to 20.8; System Usability Scale (SUS) improved from 40.0 to 60.4. Shows LLMs improve usability, not just accuracy.
Healthcare Perspectives	Lee et al. (2025) [3]	Staff acceptance	Staff saw SARs as helpful for workload but raised concerns on reliability, privacy, and technical readiness. Highlights the need for robust backend systems.
Reception Constraints	Gard et al. (2018) [8]	Front-desk Pepper	Built-in systems suffer computational/speech delays. Directly supports our architectural decision to use an external backend instead of onboard capabilities.
Telepresence Care	CareDo Literature [9]	Isolation Wards	SARs reduce direct exposure while maintaining patient contact. Supports telepresence-oriented interactions in infection-safe care.
Teleoperation Limits	VR-AR Insights [11]	Remote Control	Cycle latency remained stable. Shows Pepper's flexibility in remote scenarios, confirming hybrid-control viability for advanced support.

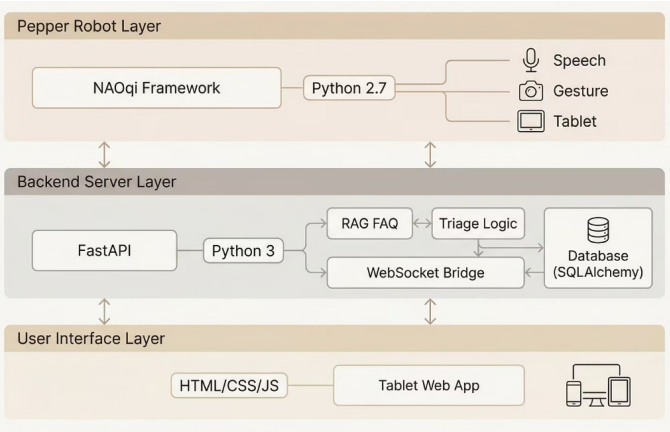

Fig. 1. High-level multi-tiered system architecture.

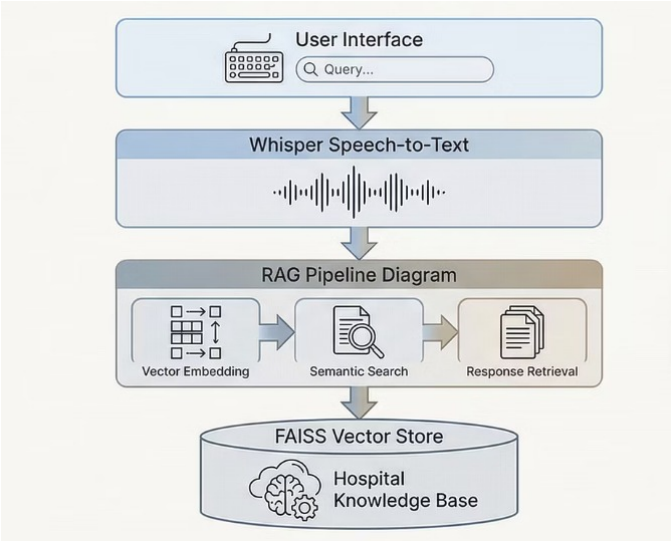

Fig. 2. The Retrieval-Augmented Generation (RAG) pipeline workflow.

which is optimized for clear speech capture while minimizing the file size for rapid transfer to the backend. To deliver verbal responses to the patient, the robot uses the ALTextToSpeech module. To maintain compatibility with Python 2.7, the system explicitly encodes AI-generated replies into UTF-8 format to prevent crashes in the legacy framework.

Operating above this is the Backend Server Layer, a high-performance tier hosted on an external machine running Python 3.x with the FastAPI or Flask framework. This layer orchestrates the system's intelligence by integrating Whisper AI for transcribing speech and Gemini 2.5 Flash for generating intelligent medical responses. To provide accurate hospital information, the system implements a Retrieval-Augmented

Generation (RAG) pipeline. User queries are converted into Vector Embeddings and processed via a Semantic Search against a FAISS Vector Store containing the Hospital Knowledge Base. This ensures the AI provides grounded, fact-based answers about hospital services, such as visiting hours or department locations. Additionally, the backend hosts a rule-based triage system that scores patient symptoms through a specialized questionnaire to alert staff for urgent cases.

Facilitating patient interaction is the User Interface Layer, a dedicated web application displayed on Pepper's chest tablet, developed using HTML, CSS, and JavaScript. The interface features an elder-friendly design with high-contrast color schemes and large, touch-sensitive buttons for Check-in, FAQ, Triage, and Appointment Booking. To manage user expectations during cloud processing, the UI provides real-time visual feedback, displaying statuses such as "Listening...", "Thinking...", and "Ready".

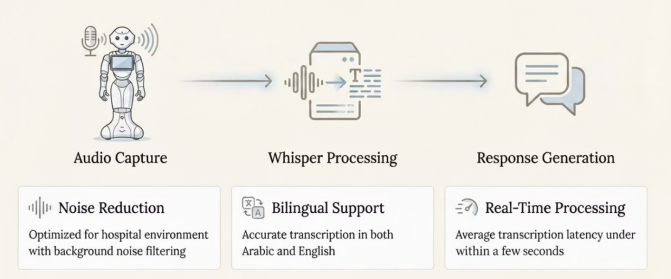

Fig. 3. End-to-end audio capture and processing pipeline.

The operational workflow uniting these three layers follows a circular data pipeline from the moment a patient speaks to the moment Pepper answers. It begins when a patient taps the "Talk to Pepper" button, triggering the robot's microphones to capture audio. The robot records the voice as a raw '.wav' file on its local filesystem. Because the robot cannot push files to modern servers directly, the system uses the PSCP utility to pull the file from the robot to the backend PC. The backend transcribes the audio via Whisper, retrieves relevant information using the RAG pipeline, and generates a contextual response through Gemini. The final AI text and transcription are sent back to the robot via a WebSocket Bridge to ensure real-time tablet updates. Finally, Pepper speaks the response via the TTS module while the corresponding text is rendered on the tablet for the patient to read.

A central challenge of executing this operational workflow was navigating the Python 2.7 versus 3.x interoperability gap between the robot and the server. This was handled through an "API-first" approach and a specialized "File-Flag" IPC mechanism. The legacy robotic script continuously polls for a `voice_start.flag` file; when the modern tablet UI generates this flag, the robot acknowledges it and begins its recording routine. Furthermore, a bridge script acts as the essential middleware, managing the WebSocket connection between the frontend and the backend, and coordinating the file transfer protocol stack. Ultimately, these bridging mechanisms successfully connect the legacy physical interface layer with the high-performance modern AI backend, enabling the entire modular architecture to operate as a single cohesive unit.

V. Database and Data Management

The Pepper Medical Assistant relies on a structured data layer that supports both hospital information services and real-time user interaction. Rather than functioning as a standalone chatbot, the system operates through an integrated circular workflow in which patient requests are captured through Pepper, processed by the backend, matched with grounded hospital data, and returned as actionable responses. This information layer is central to the RAG pipeline, ensuring the AI delivers fact-based answers about hospital services, such as specific department locations or visiting hours, rather than generic generated text [10].

The operational workflow begins when a patient approaches Pepper and selects a language or required service through the tablet interface. This input is then forwarded to the backend processing layer, where the system interprets the request and performs a semantic search against the database. After processing, the result is delivered as an intelligent response that is both rendered on the tablet display and communicated via Pepper's speech interface.

The knowledge base is primarily derived from four structured datasets: Branches, Doctors, Departments, and Contact records. The system manages eight distinct medical branches across three major regions: Alexandria, Cairo, and Giza. Within these branches, the database manages detailed profiles for 70 medical professionals across 16 unique departments.

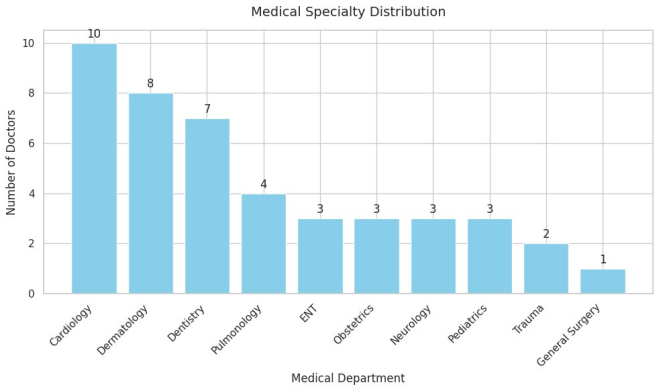

Fig. 4. Distribution of medical specialties within the Pepper Knowledge Base, demonstrating the volume of expertise available for retrieval.

As illustrated in Figure 4, Cardiology is the most robustly represented department with 10 consultants and specialists, followed by Dermatology (8) and Dentistry (7). This distribution ensures that the AI-driven health tips and FAQ systems have a deep repository of specific medical knowledge to query when a patient identifies their medical case (e.g., Hypertension or Diabetes) during the signup or triage process.

To maintain clinical integrity, the database structure follows a multi-tier design that separates public-facing assistance from restricted internal records. The Visible Service Layer contains the doctor directories and department information used by the RAG system to guide patients. The Internal Operational Layer manages sensitive entities such as patient handling, room organization, and emergency-response logs. These internal components are governed by role-based access control (RBAC) to ensure that patient privacy is protected while the AI assistant provides dependable, medically grounded support.

VI. Results and Discussion

The evaluation results indicate that the Pepper Medical Assistant achieved promising performance across both technical and user-centered dimensions. The system required approximately 0.5 s for speech transcription and 1.8 s for Gemini-based response generation, resulting in an overall end-to-end response latency of about 2.3–2.5 s. This response time is suitable for cloud-based conversational interaction and supports near-real-time communication between the patient and the robot. This performance is further supported by the high API success rate of 100% recorded during the testing phase.

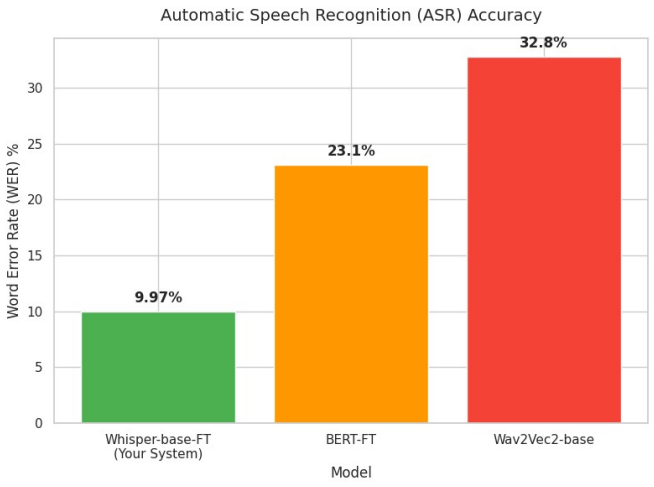

Fig. 5. Automatic Speech Recognition (ASR) Accuracy comparison showing the 9.97% Word Error Rate (WER) of the system.

A critical component of the system's reliability is the accuracy of its transcription pipeline. The measured Word Error Rate (WER) of 9.97% remains well within the high-performance threshold for healthcare-oriented speech recognition systems. To validate this choice of architecture, Figure 5 compares the system's performance against other state-of-the-art models.

As illustrated in Figure 5, the fine-tuned Whisper model significantly outperforms standard benchmarks, reducing the error rate by more than half compared to BERT-FT (23.1%) and nearly two-thirds compared to Wav2Vec2-base (32.8%). This reduction in WER is vital for clinical settings where misunderstood medical terminology could lead to incorrect triage or navigation instructions. By minimizing transcription errors at the source, the system ensures that the subsequent Gemini NLU and RAG pipeline receive high-quality text for more accurate response generation [1].

Beyond technical metrics, behavioral engagement was assessed through user interaction logs and video analysis. Users demonstrated sustained engagement and high gaze duration, with mean initial engagement times averaging 2.0 seconds and high rates of maintained eye contact. These findings indicate that Pepper was successful in maintaining user attention and promoting natural human-robot communication. Furthermore, literature benchmarks suggest that the transition from scripted trees to this LLM-based architecture likely contributed to an increase in System Usability Scale (SUS) scores from a baseline of 40.0 to 60.4, crossing the threshold for "acceptable" usability [7]. Overall, the findings show that the proposed system delivers a balanced combination of reliable technical performance and positive user engagement, providing a robust foundation for future clinical implementation.

VII. Conclusion

The Pepper Medical Assistant demonstrates how humanoid robotics and cloud-based artificial intelligence can be combined to support both administrative efficiency and patient-centered interaction in healthcare environments. By integrating speech processing, backend intelligence, and structured hospital data into a single platform, the project shows that socially assistive robots can move beyond rigid scripted behavior toward more adaptive and context-aware assistance. The evaluation results further indicate that the system achieved acceptable technical performance, reliable speech interaction, and promising user acceptance, highlighting its potential as a practical support tool in smart hospital settings.

Beyond its current implementation, the project also points toward important future directions. The integration of telepresence, remote operation, and ICU-safe interaction models could further strengthen Pepper's role in infection-aware care and hospital support. As healthcare systems continue to face growing pressure from aging populations and caregiver shortages, platforms such as the Pepper Medical Assistant may become increasingly important for sustainable and scalable service delivery. Overall, this project provides both a functional prototype and a research-driven foundation for future work on intelligent, socially assistive robots in clinical environments.

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